

Keyword Extraction Tool-17/10/25(Friday):

Report:

1. Project Overview and Objective

The goal of this project was to develop a comprehensive Keyword Extraction Tool capable of identifying the most important keywords and key phrases from a collection of text documents. To evaluate the best approach, we implemented and critically compared three distinct methodologies:

- TF-IDF (Statistical Keyword Extraction)
- RAKE (Rapid Automatic Keyphrase Extraction)
- KeyBERT (Embedding-based/Deep Learning Method)

2. Text Preprocessing and Data Pipeline

Successful keyword extraction relies on clean input. A crucial step was designing a flexible preprocessing pipeline that created two versions of the text data to satisfy the unique requirements of the algorithms:

1. Version 1 (With Stopwords): Used as input for RAKE and KeyBERT. These methods rely on context and delimiters (stopwords/punctuation) to correctly form multi-word phrases.
2. Version 2 (Without Stopwords): Used as input for TF-IDF. Removing common words ensures that frequency counts are focused exclusively on content words, improving scoring accuracy.

All text was tokenized, lowercased, and lemmatized prior to extraction.

3. Visualization of Corpus Keywords

The following visualizations, generated from the fully preprocessed text (TF-IDF input), summarize the overall frequency and importance of keywords in the corpus.

(The visualization code generates the images; you would replace the bracketed text below with the actual output images.)

Word Cloud of Top Content Words

Top 10 Keyword Frequency Distribution

(Bar Chart)

4. Final Evaluation and Method Comparison

The project concludes by evaluating the output quality of all three methods, focusing on Interpretability, Accuracy, and Speed.

A. Summary Table: Comparing Extraction Methods

Criterion	TF-IDF (Statistical)	RAKE (Heuristic/Phrase)	KeyBERT (Embedding)		
Output Format	Single Keywords	Long, Descriptive Phrases	Balanced, Contextual Phrases		
Interpretability	Fair (Lacks context; concepts often fragmented)	Good (Specific, readable phrases)	Excellent (Most concise and clear)		
Accuracy/Relevance	Fair (Sensitive to common words)	Good (Effective for descriptive language)	Excellent (Best contextual fit using BERT)		
Computational Speed	Fastest (Simple matrix math)	Very Fast (Rule- based)	Slowest (Requires large model loading/embeddings)		
Complexity	Low	Low	High		

B. Sample Document Comparison

The sample outputs below demonstrate the functional differences, where \$N=3\$ top results were extracted:

Document Sample	TF-IDF Keywords	RAKE Keyphrases	KeyBERT Keyphrases		
D1 (Gemini AI)	working, google, intelligence	new artificial intelligence model called gemini	model called gemini, new artificial intelligence		
D3 (Power Gen.)	power, generation, suggests	solar power generation, coal power generation	cheaper than coal, solar power generation		
D4 (Profits)	profit, report, rising	financial report detailed rising profit	detailed rising profit, rising profit for		

5. Conclusion

The analysis clearly shows a trade-off between efficiency and quality:

1. Efficiency: TF-IDF is the undisputed winner on speed, making it suitable for indexing massive datasets where high precision is not required.

2. Quality: KeyBERT provided the highest quality, most accurate, and most interpretable phrases by leveraging deep learning's ability to understand context.
3. Balance: RAKE is an excellent compromise, delivering highly descriptive multi-word phrases nearly as fast as TF-IDF, making it ideal for applications requiring quick, human-readable summaries.

The project concludes that for applications prioritizing semantic accuracy and interpretability (like content tagging), KeyBERT is the superior solution. For high-volume, real-time use cases, RAKE offers the best performance-to-quality ratio.

SUMMARY:

This project successfully developed and evaluated a Keyword Extraction Tool by implementing three distinct methodologies: TF-IDF (statistical), RAKE (heuristic), and KeyBERT (embedding-based).

The primary objective was to determine the most effective approach for extracting representative keywords and key phrases from text.

Key Findings:

- RAKE and KeyBERT provided superior output quality by extracting multi-word key phrases (e.g., "solar power generation"), which are more contextually relevant than the single words extracted by TF-IDF.
- KeyBERT demonstrated the highest overall accuracy and interpretability due to its ability to understand semantic context using BERT embeddings, making its phrases the most precise.
- TF-IDF was the fastest method, proving it's best suited for high-volume, initial indexing tasks where speed is prioritized over semantic depth.

Conclusion: For high-quality content summarization and tagging, KeyBERT is the most effective solution. For applications requiring rapid, descriptive output, RAKE offers the best balance of speed and relevant phrase extraction.